



KOLEJ MATRIKULASI KEJURUTERAAN JOHOR

JOHOR MATRICULATION TECHNICAL COLLEGE

PEPERIKSAAN SEMESTER PROGRAM MATRIKULASI

MATRICULATION PROGRAMME EXAMINATION

MATEMATIK 1

MATHEMATICS 1

2 JAM

2 HOURS

LIST OF MATHEMATICAL FORMULAE

SENARAI RUMUS MATEMATIK

Quadratic equation $ax^2 + bx + c = 0$:

Persamaan kuadratik $ax^2 + bx + c = 0$:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Trigonometry**Trigonometri**

$$\sin (A \pm B) = \sin A \cos B \pm \cos A \sin B$$

$$\cos (A \pm B) = \cos A \cos B \mp \sin A \sin B$$

$$\tan (A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$$

$$\sin A + \sin B = 2 \sin \frac{A+B}{2} \cos \frac{A-B}{2}$$

$$\sin A - \sin B = 2 \cos \frac{A+B}{2} \sin \frac{A-B}{2}$$

$$\cos A + \cos B = 2 \cos \frac{A+B}{2} \cos \frac{A-B}{2}$$

$$\cos A - \cos B = -2 \sin \frac{A+B}{2} \sin \frac{A-B}{2}$$

$$\sin 2A = 2 \sin A \cos A$$

$$\cos 2A = \cos^2 A - \sin^2 A$$

$$= 2 \cos^2 A - 1$$

$$= 1 - 2 \sin^2 A$$

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

$$\sin^2 A = \frac{1 - \cos 2A}{2}$$

$$\cos^2 A = \frac{1 + \cos 2A}{2}$$

Differentiation

Pembezaan

$f(x)$	$f'(x)$
$\cot x$	$-\operatorname{cosec}^2 x$
$\sec x$	$\sec x \tan x$
$\operatorname{cosec} x$	$-\operatorname{cosec} x \cot x$

$$\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx} \qquad \frac{d^2y}{dx^2} = \frac{\frac{d}{dt} \left(\frac{dy}{dx} \right)}{\frac{dx}{dt}}$$

Sphere

Sfera

$$V = \frac{4}{3} \pi r^3$$

$$S = 4 \pi r^2$$

Right circular cone

Kon membulat tegak

$$V = \frac{1}{3} \pi r^2 h$$

$$S = \pi r^2 + \pi r h$$

Right circular cylinder

Silinder membulat tegak

$$V = \pi r^2 h$$

$$S = 2\pi r^2 + 2\pi r h$$

Section A (25 marks): Answer all questions in this section.

1. Find the limit of the following, if it exist

(a) $\lim_{x \rightarrow 2} \frac{(2-x)^2}{4-2x}$.

[3 marks]

(b) $\lim_{x \rightarrow -\infty} \frac{\sqrt{x^2 + 6x}}{3 - 2x}$

[3 marks]

(c) $\lim_{x \rightarrow 1} \frac{x^2 - 3x + 2}{\sqrt{x} - 1}$

[3 marks]

2. (a) By using the first principle of differentiation, find $y = \sqrt{8x-3}$.

[5 marks]

- (b) Find the derivative of the following function:

(i) $y = \ln\left(\frac{x-5}{x+3}\right)$

[3 marks]

(ii) $y = \cos(e^{2x})$

[3 marks]

3. Given that $f(x) = 3x^4 - 4x^3 + 1$. Find the stationary point by using the first derivative test. Hence determine the relative maximum or minimum (if any)

[5 marks]

Section B (75 marks): Answer all questions in this section.

1. Given that $p = 1 - i$. Express $z = \frac{p+2-i}{p^2-i}$ in the form of $a + bi$, where a and b are real number. Hence express z in polar form.
[9 marks]
2. (a) Determine the solution set of inequality $\left| \frac{x}{5+x} \right| > 2$
[7 marks]
- (b) Find the exact value of x for $\frac{4}{2+e^x} = \frac{6}{3+e^{-x}}$.
[5 marks]
3. Let $f(x) = \ln(2x+3)$ and $g(x) = e^{-x} + 3$ be two functions.
- (a) Find $(g \circ f)^{-1}(x)$. Hence, calculate $(g \circ f)^{-1}(2)$.
[6 marks]
- (b) Find $g^{-1}(x)$. Hence, determine the value of x if $g^{-1} \circ f(x) = 0$.
[5 marks]
4. A function g is defined as $g(x) = \frac{1}{3} \ln\left(\frac{x+1}{2}\right)$ for $x > -1$. Determine the inverse function of g and hence, sketch the graphs of g and g^{-1} on the same axes. Hence, state the domain and range for g^{-1} .
[7 marks]

5. (a) Given the function

$$f(x) = \begin{cases} \frac{4}{x^2} + p & , x < -2 \\ (4+p)x^2 & , x \geq -2 \end{cases}$$

Determine the value of p if $\lim_{x \rightarrow -2} f(x)$ exists. State the value of $f(-2)$ so that $f(x)$ is continuous at $x = -2$.

[5 marks]

- (b) The function $g(x)$ is given by

$$g(x) = \begin{cases} \frac{e^{2x} - 49}{e^x - 7}, & x \in \mathbb{R}, x \neq \ln 7 \\ 12, & x = \ln 7 \end{cases}$$

Determine whether $g(x)$ is continuous at $x = \ln 7$.

[6 marks]

6. (a) Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$, given that $y^2 + xy - 3x = 7$

[9 marks]

- (b) Find the first and second derivative of the function defined by the parametric equation.

$$x = 3e^{2t}, \quad y = 3^{3t} - 1$$

[7 marks]

7. A cylindrical container which is open at the top and height h meter and radius r meter has a capacity of $5m^3$. Find the value of r which makes the area as small as possible.

[9 marks]

END OF QUESTION PAPER

SUGGESTED FINAL ANSWER

Section A

1. (a) 0 (b) $\frac{1}{2}$ (c) -2

2. (a) $\frac{dy}{dx} = \frac{4}{\sqrt{8x-3}}$; (b)(i) $\frac{dy}{dx} = \frac{1}{x-5} - \frac{1}{x+3}$ (ii) $\frac{dy}{dx} = -2e^{2x} \sin(e^{2x})$

3. (1,0) is the relative minimum point

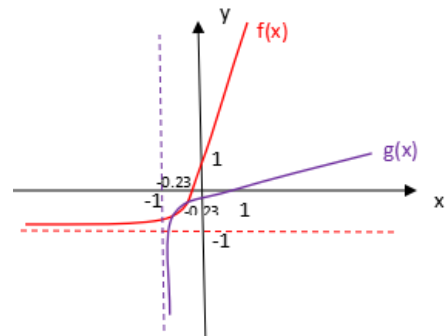
Section B

1. $z = \frac{2}{3} + i, z = \frac{\sqrt{13}}{3} [\cos(0.983\text{rad}) + i \sin(0.983\text{rad})]$

2. (a) $\left\{x: -10 < x < -5 \text{ or } -5 < x < -\frac{10}{3}\right\}$; (b) $x = \frac{1}{2} \ln \frac{2}{3}$

3. (a) $\frac{10-3x}{2x-6}; -2$ (b) $-\ln(x-3); \frac{e^4-3}{2}$

4. $g^{-1}(x) = 2e^{3x} - 1; D_g = R_{g^{-1}} = (-1, \infty), R_g = D_{g^{-1}} = \mathbb{R};$



5. (a) $p = -5, f(-2) = -4$

(b) $f(x) \neq \lim_{x \rightarrow \ln 7} f(x)$, not continuous at $x = \ln 7$

6. (a) $\frac{dy}{dx} = \frac{3-y}{2y+x}; \frac{d^2y}{dx^2} = \frac{(y-3)(2y+2x+6)}{(2y+x)^3}$ (b) $\frac{dy}{dx} = \frac{3^{3t} \ln 3}{2e^{2t}};$

7. $r = 1.17m$